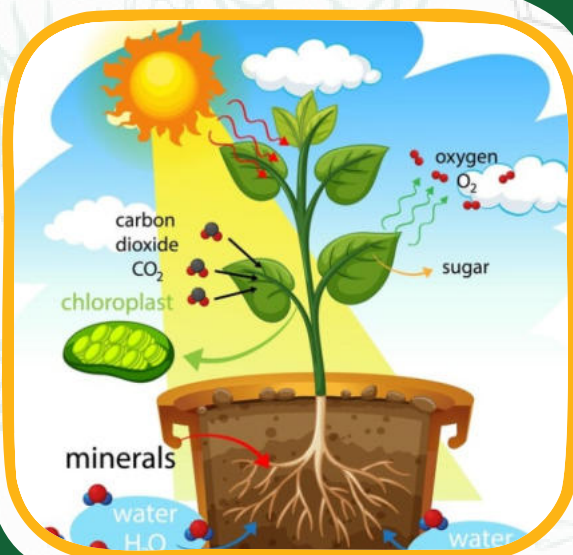
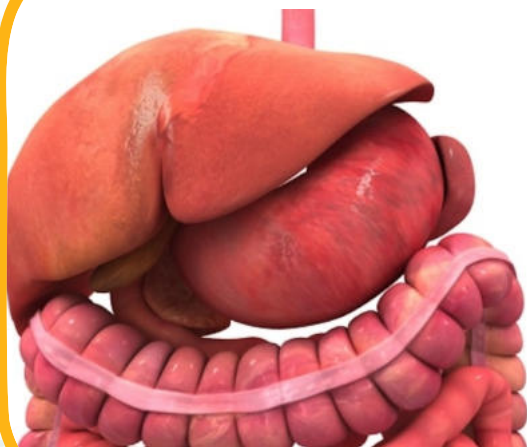
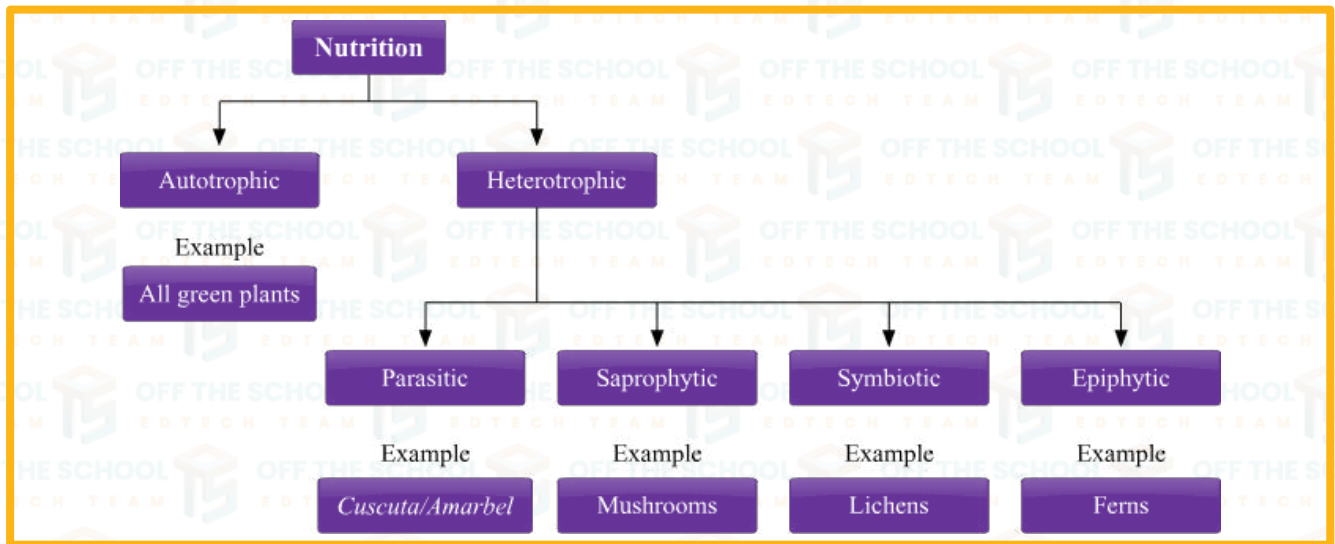




NUTRITION



**Definitions:****NUTRITION:**

Process by which organisms obtain and use the nutrients required for maintaining life is called nutrition.

NUTRIENTS:

Essentials substances that our body needs in order to grow and stay healthy are known as nutrients.

MALNUTRITION:

A condition that results from taking an unbalanced diet either taking or in excess of certain nutrients.

OBESITY:

Obesity means becoming overweight and it may also be due to malnutrition.

Ingestion:

The intake of food is known as ingestion.

CHYME:

Food that stays in your stomach for a few hours gradually turns into a thick semi-fluid substance called chyme.

STARVATION:

It is a severe reduction in nutrients and energy intake and is the most horrible effect of malnutrition.

MODES OF NUTRITION

Living organisms can be divided into two groups on the basis of their mode of nutrition.

1. Autotrophic organisms.

2. Heterotrophic

AUTOTROPHIC ORGANISM:**Definition:**

In this mode of nutrition in which organisms can prepare their own food from inorganic raw material taken from their surroundings are known as autotrophic organisms.

Example:

Autotrophs are photosynthetic like plants, algae and some bacteria etc. Heterotrophic Organism

Definition:

In this mode of nutrition organism which are unable to prepare their own food from Simple inorganic compounds are known as heterotrophic organisms.

Example:

Most bacteria, fungi and animals have heterotrophic modes of nutrition.

NUTRITION IN PLANTS**AUTOTROPHIC NUTRITION:****Definition:**

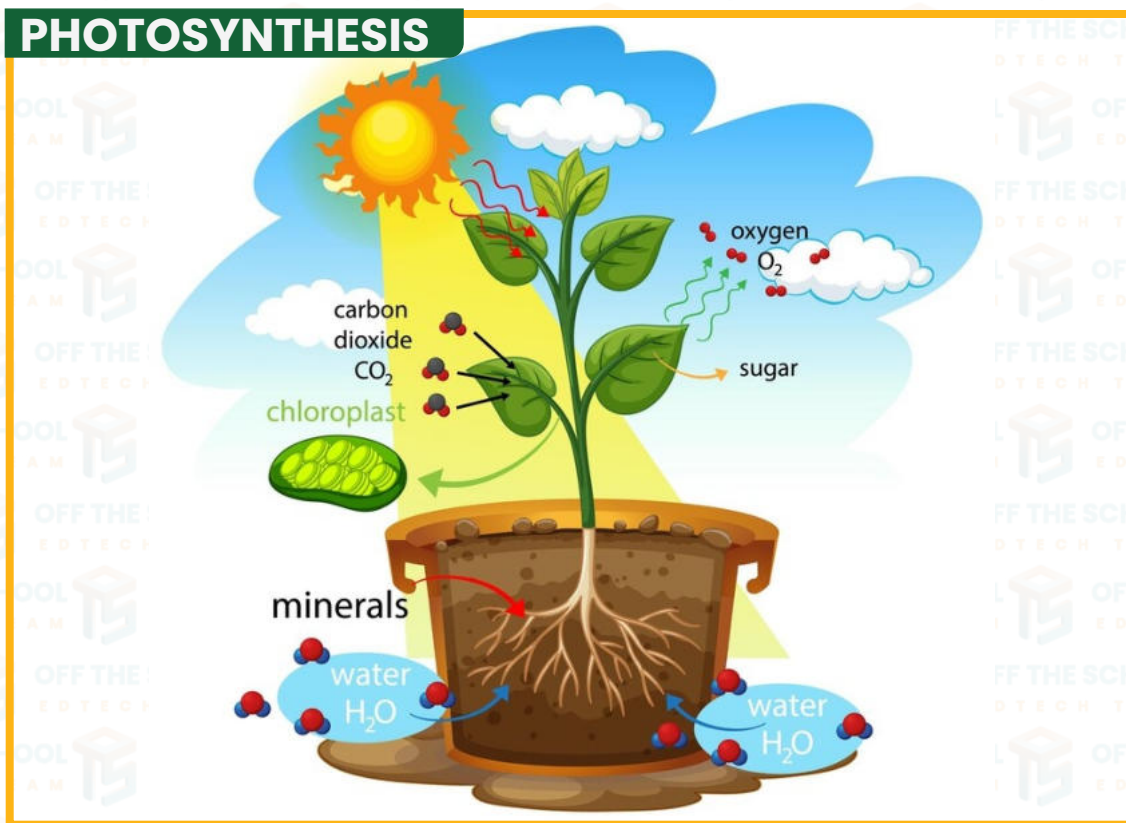
AUTOS: (self) and **TROPHE:** (nutrition) In this mode of nutrition organisms which can prepare their own food from inorganic raw material such as carbon dioxide and water taken from their surroundings are known as autotrophic organisms.

Examples of Autotrophic modes of nutrition:

In this mode of nutrition in which organisms can prepare their own food from inorganic raw material taken from their surroundings are known as autotrophic organisms.

PHOTOSYNTHESIS:**Definition:**

The process by which plants manufacture their organic compounds by taking inorganic compounds from their surroundings, in the presence of sunlight is called photosynthesis.

Procedure:**PHOTOSYNTHESIS**

Sunlight provides the energy which is stored in chlorophyll of the plant cells. Enzymes are also needed for this process. This process is completed in the mesophyll cells of the leaves where chlorophyll is present. In this way the plant builds sugar molecules from simple compounds i.e. carbon, oxygen and hydrogen.

Chemical Equation:

HETEROTROPHIC NUTRITION:

Definition:

HETEROTROPHIC: In this mode of nutrition in which organisms cannot synthesis their own food from simple inorganic materials such as carbon dioxide, water and depend on other organisms from its food.

Consumer:

The animals and all other organisms that cannot make their own food area called consumers.

TYPES OF CONSUMERS:

There are three main types of consumers.

1. Herbivores
2. Carnivores
3. Omnivores

HERBIVORES:

Herbivores are the animals that eat plants. Goat, cow etc.

CARNIVORES:

Carnivores are animals that eat other animals. Lion, leopard etc.

OMNIVORES:

Omnivores are the animals that eat both plants and animals e.g., human, monkey etc.

Types of Heterotrophic Nutrition

Heterotrophic mode of nutrition is of three types.

1. Parasitic Nutrition
2. Saprotrophic
3. Holozoic

PARASITIC NUTRITION:

They entirely or partially depend upon autotrophic plants for their food supply they grow on the stem of the host plant.

Example:

1. Cuscuta
2. Copepods
3. Lac insect
4. Leeches

Cuscuta



Copepods



Lac insect



Leeches



Saprotrophic:

An organism obtains its food from dead and dying organic matter.

Example:

1. Fungi (Moulds, Mushrooms & Yeasts)
2. Many Bacteria

Holozoic:

An organism has to obtain organic molecules from their environment because they cannot make their food from simple inorganic compounds. This type of nutrition is known as holozoic nutrition.

Example:

- Non-Parasitic animals (like amoeba)
- Complex organism (Human being)

Nutrition in amoeba:

Nutrition in unicellular organisms like Amoeba is known as holozoic nutrition. It involves ingestion by the cell surface, digestion and egestion. Amoeba takes in complex organic matter as food.

Ingestion by the cell surface:

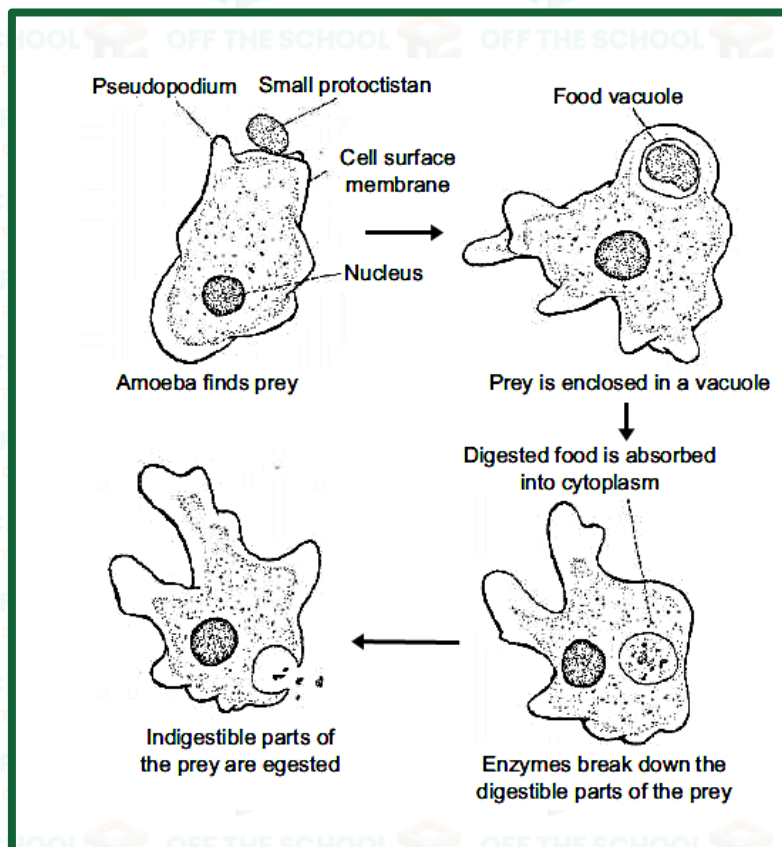
Amoeba first identifies its food then throws out a number of small pseudopodia (projections of cytoplasm, also called false feet). These pseudopodia enclose the food particle and prevent it from escaping. The food enclosed in the cell membrane forms a food vacuole.

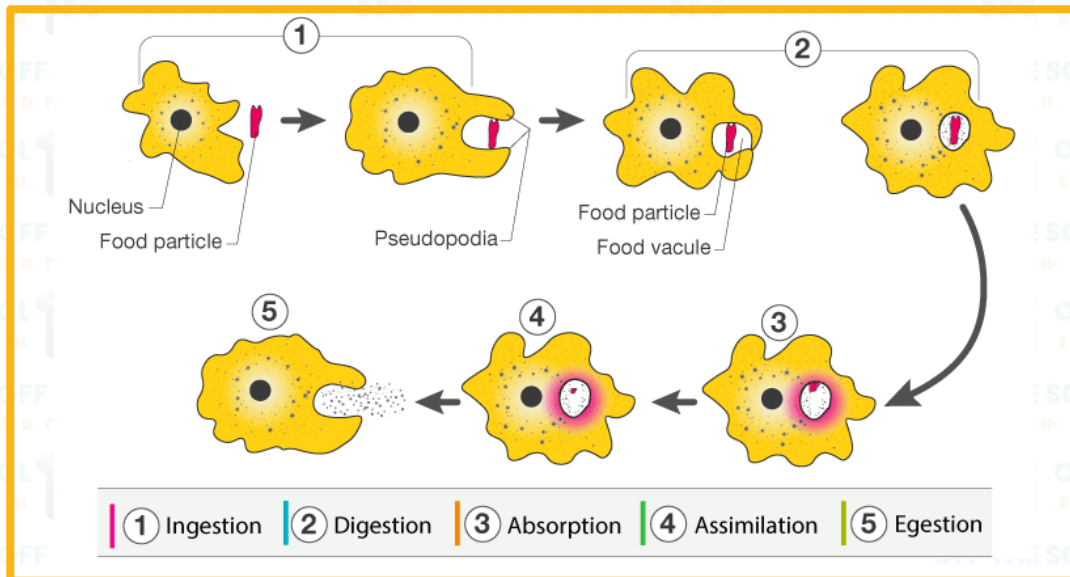
Digestion:

The complex food is broken down into simpler molecules with the help of digestive enzymes produced by an organelle called lysosome. The digested food is distributed in the cytoplasm.

Egestion:

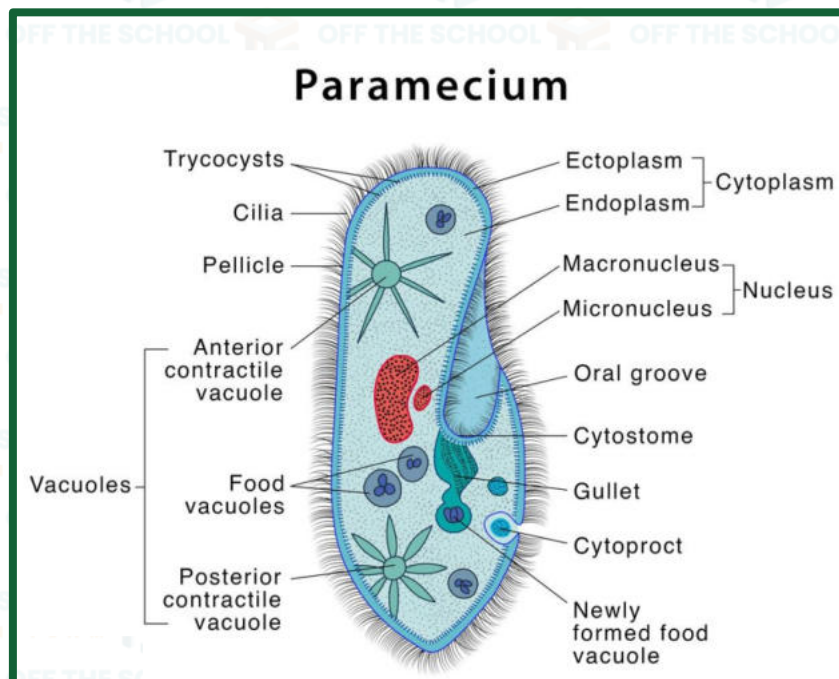
The undigested food is egested through the cell membrane.





Nutrition In Paramecium:

In Paramecium, a unicellular organism with a specific shape, food is ingested through a special opening, the cytostome (cell mouth). Food is brought to this opening by the lashing movement of cilia that cover the entire surface of the cell.



Mineral Nutrition in Plants

Definition:

The process involving the absorption, distribution and utilization of mineral substances by the plants for their growth and development is called mineral nutrition.

Mode of Nutrition:

Plants have the most efficient mechanisms for autotrophic mode of nutrition.

Types of Nutrition:

There are two types of nutrients.

1. Macronutrients

2. Micronutrients

Macronutrients:

The nutrients which are required by plants in large quantities are called macronutrients.

Example:

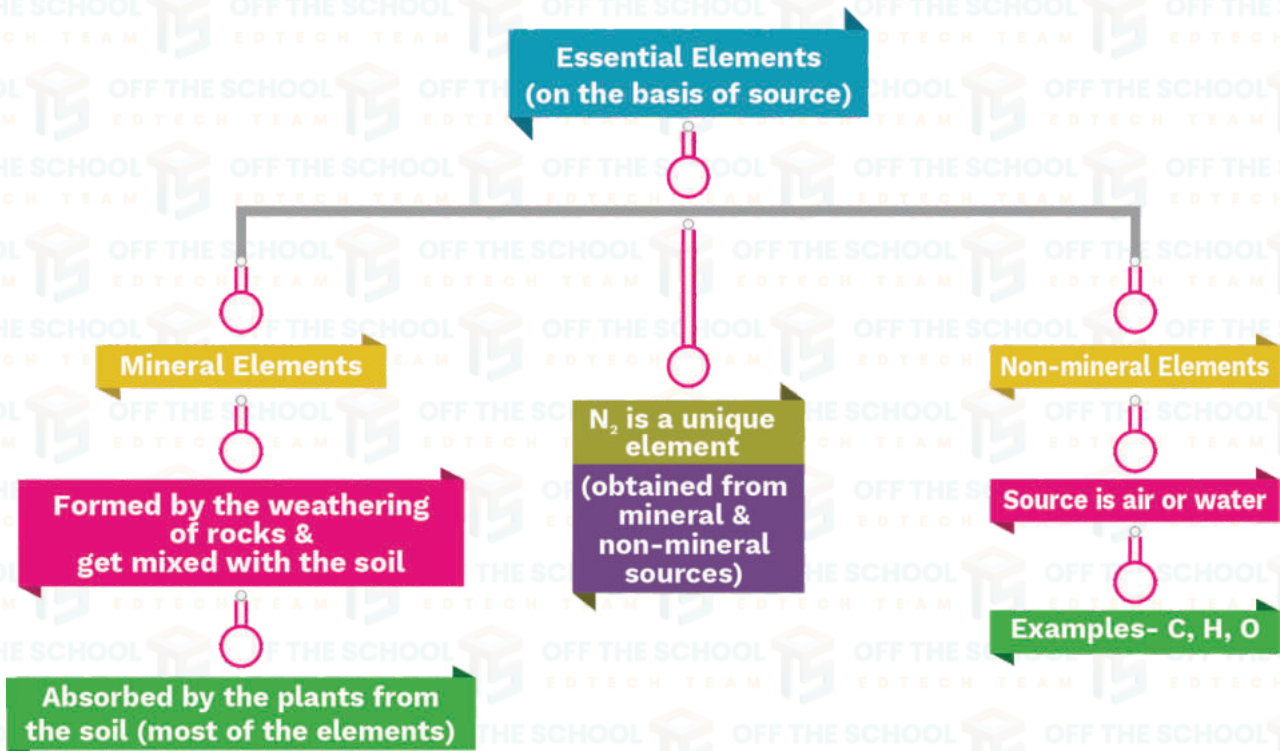
Carbon, Hydrogen, Oxygen, Nitrogen, Magnesium, Potassium etc.

Micronutrients:

The nutrients which are required by plants in small quantities are called micronutrients.

Example:

Iron, Molybdenum, Boron, Chlorine, Zinc etc.

**Role of Nitrogen in Plants:**

Plants get nitrogen in the form of nitrates. Nitrogen metabolism is a major factor in stem and leaf growth. It helps plants with rapid growth, increasing seed and fruit production and improving the quality of leaf and crops.

Major Components:

Nitrogen is a major component of the following which are highly essential for plant life.

Hormones.

Chlorophyll

Vitamins

Enzymes

Excessive Nitrogen:

Too much nitrogen can delay flowering & fruiting.

Deficiency:

Deficiency of nitrogen can reduce yields

Cause yellowing of leaves

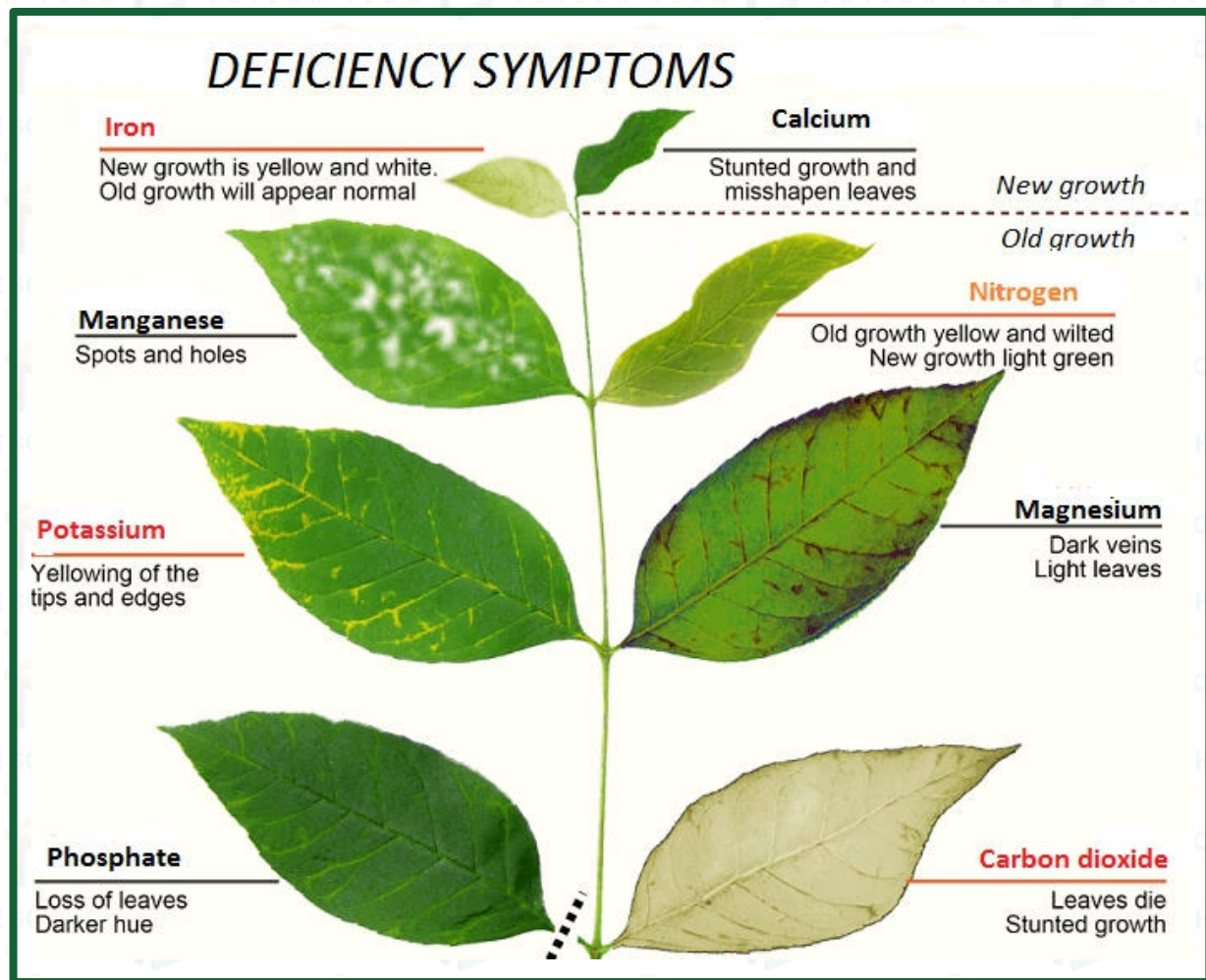
Stunt growth

Role of Magnesium in Plants:

Magnesium is present in the soil in water-soluble exchangeable and fixed form and it is present in many primary minerals. It is essential for the synthesis of fats and the metabolism of Carbohydrates. It is also essential for fruit and nut formation & germination of seeds.

Deficiency:

Deficiency of magnesium causes yellowing and wilting of leaves.



Main Components of Human Food

The following are the seven major classes of nutrients

1. Carbohydrates
2. Protein
3. Minerals
4. Fiber
5. Vitamin
6. Water

1. Carbohydrates:

These are organic compound which mostly contain elements such as Carbon, hydrogen and oxygen. Each molecule of carbohydrates contain hydrogen and oxygen in a ratio of 2:1.

Simple Carbohydrates:

Glucose Grape sugar

Fructose Fruits sugar

Complex Carbohydrates:

1. Sucrose
2. Cellulose
3. Starch

Source of Carbohydrates:

Animals obtain carbohydrates from their diet rich in carbohydrates. Many vegetables, cereals & milk are rich in carbohydrate plants.

Importance of Carbohydrates:

Carbohydrates our body's first source of energy in the body.

Complex carbohydrates first broken down.

2. Proteins:

Proteins are the large Complex organic molecules containing carbon, hydrogen, nitrogen, Oxygen and Sulphur as their main constituents and are composed of one or more long chains of amino and called polypeptide chains.

Types of Amino Acid:

There are 20 different amino acid are together through peptide bond in different arrangements to produce different polypeptide chains or proteins.

Amino acids are of two types.

1. Essential amino acids
2. Non-essential amino acids

Sources of Proteins:

Food rich in protein include red meat, fish, poultry, dairy products, nuts and cereals.



Importance of Proteins:

1. Their main role is to serve as building block material in the body.
2. They help in growth & repair of body parts.
3. Proteins are essential for growth & development of an organism.
4. They make structures like muscles, nails and hair.
5. The body needs protein to maintain & replace tissue and their function.



3. Fats:

Fats are organic compounds which are mostly made up of smaller chemical units, such as fatty acids and one glycerol molecule. They are highly energy efficient foods.

Types of Fatty Acids:

The fatty acids present in lipids are of two types.

Saturated Fatty Acid:

Saturated fatty acids have all of their carbon atoms bonded to hydrogen atoms. Lipids containing saturated fatty acids are solid at room temperature.

Example:

Butter contains 70% saturated and 30% unsaturated fatty acids.

Unsaturated Fatty Acid:

Unsaturated fatty acids have some of their carbon atoms double-bonded in place of a hydrogen atom. Lipids containing consternated fatty acids are lipids at room temperature.

Example:

Sunflower oil contains 75% unsaturated fatty acids

Functions:

1. Fats provide insulation to the body from temperature changes and shocks.
2. Fats are used to synthesise hormones and other substances needed for the body activities.
3. Fats are stored in body parts as a source of energy in case of emergency.

Sources of Fats:

Important resources of lipids included milk, butter, cheese, eggs, mutton, fish, mustard, seeds, coconut, and dry fruits



4. Vitamin:

Vitamin is an organic molecule, an essential micronutrient that an organism's needs in small quantities for the proper functioning of its metabolism.

They are mainly classified into the following two categories.

1. Fat soluble vitamin.

2. Water soluble vitamin

1. Fat-Soluble Vitamin:

Vitamins which are soluble in organic solvents are called Fat-Soluble vitamins (A, D, E & K) are less excreted from the body as compared to water-soluble vitamins.

2. Water-Soluble Vitamin:

Vitamins which are soluble in water are called water soluble vitamins, these are vitamins B and C cooking or heating destroys the water-soluble vitamins more readily than the fat-soluble vitamin.

5. Minerals:

Minerals are inorganic elements that originate in the earth & cannot be made in the body.

Function:

They play roles in various body functions & are necessary to maintain health.

Sources of Minerals:

Most of the minerals in human's diets came directly from plants, water and animal foods.

6. Water:

Dietary Fibers: Water is an essential constituent of cytoplasm. It makes up nearly 70% of total body mass. Various substances are circulated in the body as a watery solution in the blood. Water helps in the absorption and transportation of the digested food. It helps plants for intake of minerals from the soil and in the movement of food for photosynthesis we should take sufficient quantities of water because a lot of water is lost from the body by sweating, urinating and breathing.

Dietary fibers are designed as a digestible portion of food obtained from plant soluble or insoluble taken by the organism.

Common sources of dietary fibers in our diet are leguminous plants, seed husk, whole grain food, and wheat and corn barn.

1. Fibers play a very important role in preventing constipation.

2. Soluble fibers help in lowering the blood cholesterol & blood sugar level.

3. Insoluble fibers facilitate peristaltic movement in the gut & hence promote ease in defecation.

4. Dietary fibers also form bulk of the food by absorbing water.

7. Role of Calcium:

Function:

i. Calcium is essential for the development & maintenance of bone and teeth.

ii. It is needed for maintaining cell membranes & connective tissues.

iii. It is needed for activation of several enzymes.

iv. It also aids in blood clotting.

v. Good calcium nutrition along with low salt and high potassium intake, prevents hypertension & kidney stones.

Sources:

Humans get calcium from milk, cheese, egg, yolk, beans, nuts, cabbage.



Deficiency:

1. Chest pain
2. Heart failure
3. Bones become soft
4. Blood clots slowly

8. Role of Iron

Function:

- Iron plays a major role in oxygen transport & storage.
- It is a component of hemoglobin in red blood cells.
- It is a component of myoglobin in muscle cells.
- Cellular energy production also requires iron.
- It also supports immune-function.

Sources:

Humans get iron from red meat, egg, yolk, whole wheat, spinach, mustard.

Deficiency:

Its deficiency is the most common deficiency worldwide. Iron deficiency causes anemia.

1. Pale skin
2. Poor appetite infants
3. Weakness
4. Extreme fatigue
5. Fast heart beat or shortness of breath

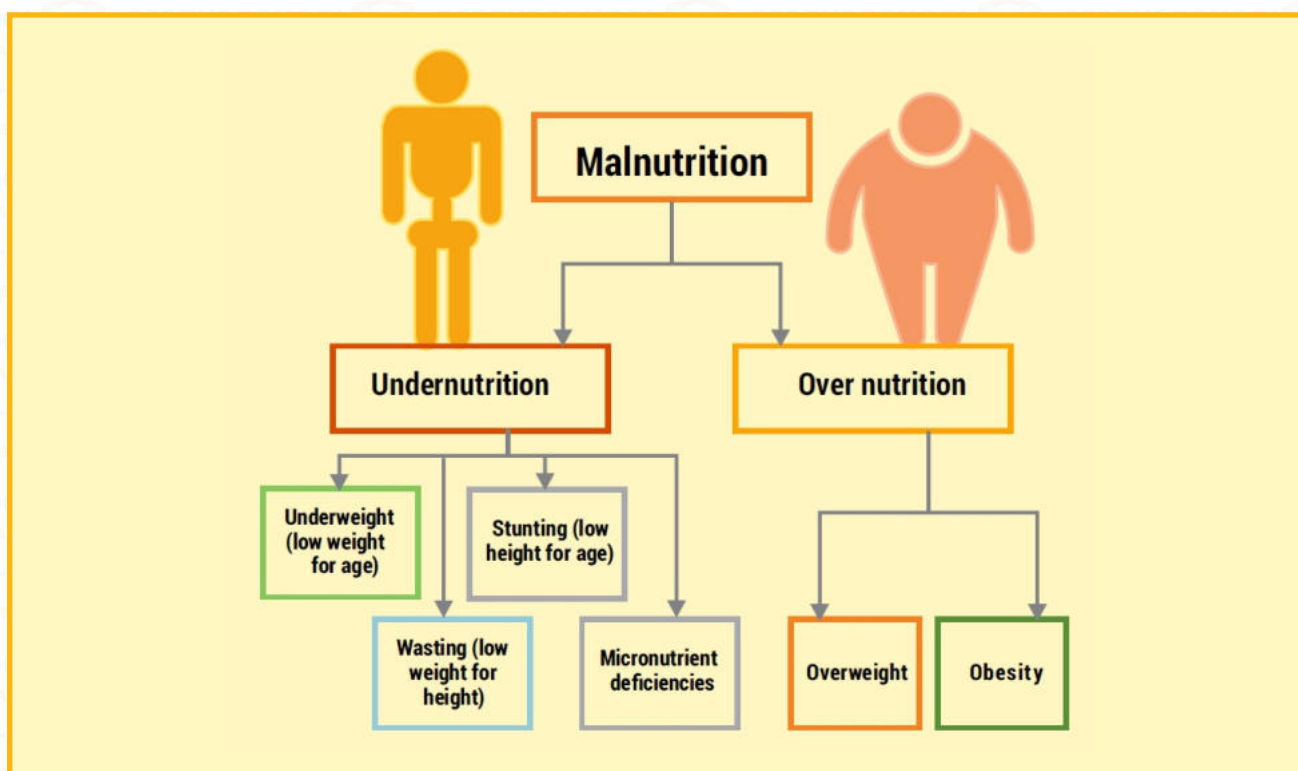
Malnutrition

If one or more basic nutrients are missing from diet. This condition is known as malnutrition.

Types of Malnutrition:

There are types of malnutrition.

1. Chronic Malnutrition
2. Acute Malnutrition



Chronic Malnutrition:

Characterized by delayed growth in the children.

Acute Malnutrition:

Characterized by insufficient weight in relation to the child's height. Acute malnutrition can be moderate or severe according to the child's height.

Three Main Nutrition Deficiency Disorder:**Vitamin A Deficiency:**

Under-nourishment & malnutrition have serious consequences for the health of the young worldwide, three main nutrition deficiencies are of particular concern.

Vitamin B Deficiency:

In the world's most common causes of preventable child blindness and vision impairment.

Iodine Deficiency:

It is the major preventable cause of mental retardation worldwide.

Protein-Energy Deficiency:

Protein-energy malnutrition means inadequate malnutrition on absorption of energy and proteins in the body. It leads to the following diseases.

Kwashiorkor**Cause:**

It is due to protein deficiency.

Age:

It occurs at the age of 12 months, usually when breast feeding is discontinued. It can also develop any time during a child's growing years.

Effects:

Children may grow to a normal height but are abnormally thin.

Symptoms:

1. Loss of muscles mass.
2. An enlarged tummy
3. Dry, brittle hair that falls out easily & may lose its colour.

Marasmus:**Cause:**

Nutrient deficiency is the main cause of marasmus.

Age:

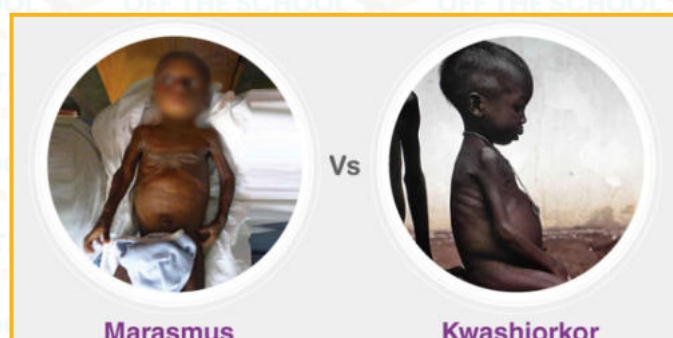
Marasmus usually develops between the ages of six months and one year.

Effects:

Patients lose all their body fat & muscle strength and acquire a skeletal appearance.

Symptoms:

1. Mood Change (always irritable)
2. Growth failure
3. Body weight is reduced by 62% of the normal wasting of muscles and subcutaneous fats.



Minerals Deficiency Diseases:

Following are the examples of disease resulting from the deficiency of a mineral.

Goiter:

A goiter is an abnormal enlargement of the thyroid gland.

Cause:

It is a condition caused by an insufficient amount of iodine in the diet.

Function of Iodine:

Iodine is used by the thyroid gland to produce hormones that control the body's normal functioning and growth.

Anemia:

The term anemia literally means "a lack of blood". The means condition is caused when the number of red blood cells reduced to a level lower than normal.

Cause:

Deficiency of hemoglobin due to the lack of iron is one of the main causes of anemia.

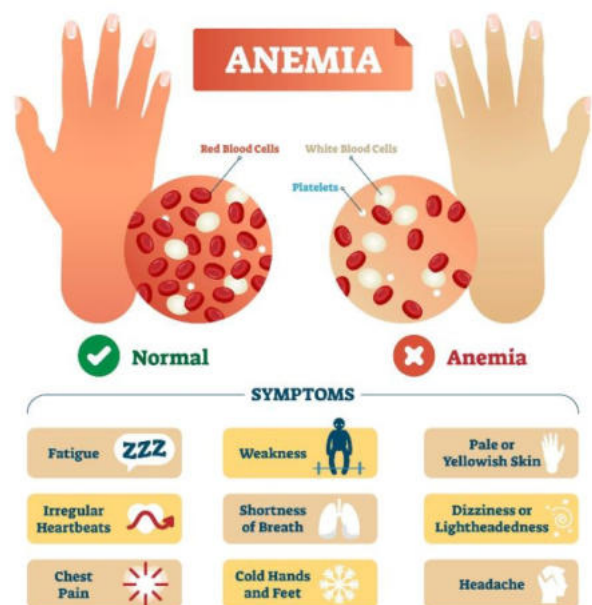
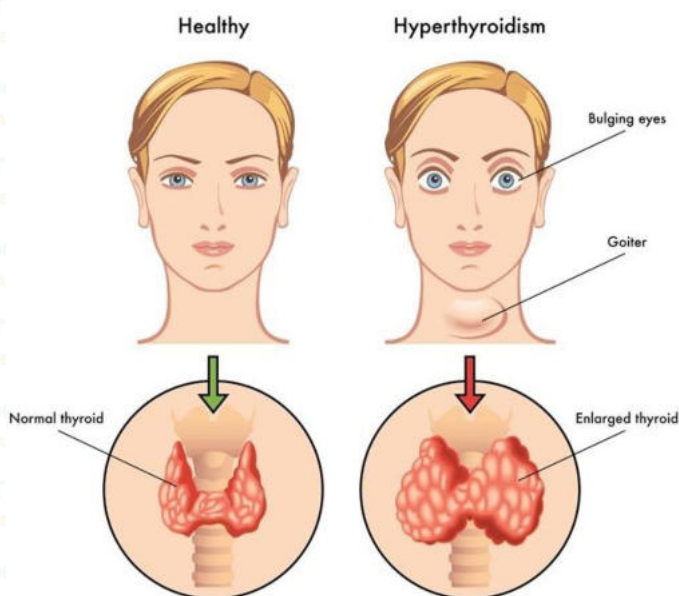
Symptoms:

1. Weakness
2. Fatigue
3. Cold hand and feet
4. Shortness of breath
5. When a person becomes weak there is a shortage of oxygen supply to the body's cells.

Over Intake of Nutrients:

Over intake of nutrients (OIN) is form of malnutrition in which more nutrients are taken than the amounts required for normal growth, development and metabolism. The effects of over intake of nutrition are usually intensified when there is reduction in daily physical activity. Also, a cause of a number of health problems. For example, high intake of carbohydrates and fats leads to obesity, Diabetes and cardiovascular problems.

- High doses of vitamins cause loss of appetite, liver problems.
- Excessive intake of Vitamin D leads to deposition of calcium in various tissues.



Human Digestive System

It is a system which comprises a complex series of various organs and digestive glands which work in coordinate with each other to carry out a function of the digestion of food.

Digestion:

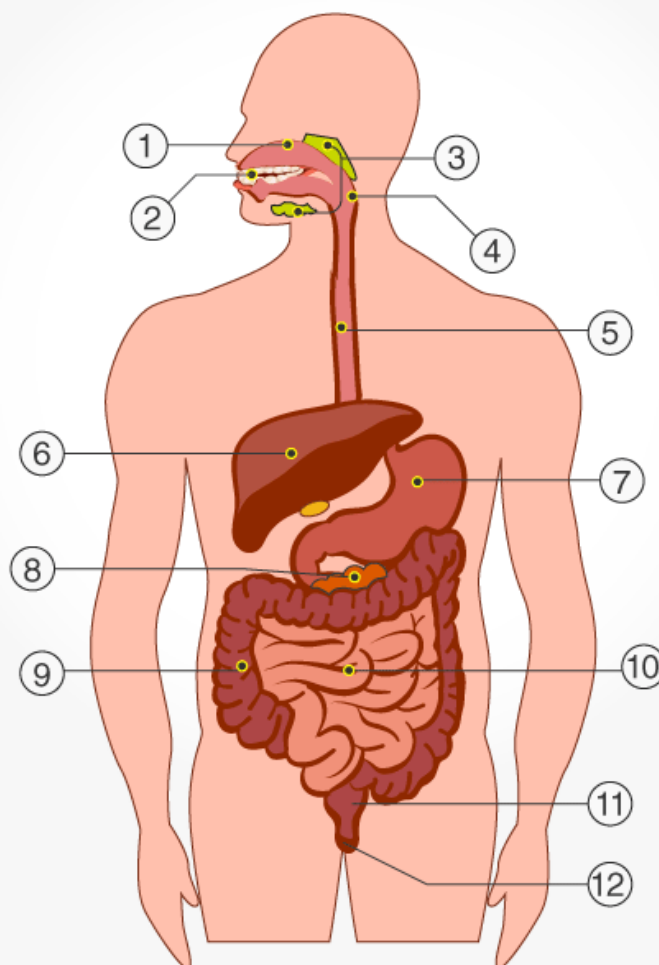
Digestion is a process in which large and non-diffusible molecules of food are converted into smaller and diffusible molecules by the action of enzymes.

Digestive system consists of two parts.

- Alimentary canal
- Digestive glands

Alimentary Canal:

The alimentary canal is divided into



1 Mouth	2 Teeth	3 Salivary glands	4 Pharynx
5 Esophagus	6 Liver	7 Stomach	8 Pancreas
9 Large intestine	10 Small Intestine	11 Rectum	12 Anus

- Oral cavity
- Pharynx
- Oesophagus
- Stomach
- Small intestine
- Large intestine

Oral Cavity:

It is the first part of the alimentary canal. It is equipped with four kinds of teeth. Oral cavity or mouth also contains three pairs of ordinary glands which produce saliva.

Function:

The function oral cavity is given below

Food Selection:

In oral cavity food is selected or rejected due to taste, hard object or diet.

Grinding of Food:

The second function of the oral cavity is the grinding of food by teeth. It is known as chewing or mastication. Grinding increases the surface area of food.

Lubrication of Food:

The salivary glands of the oral cavity secrete which lubricates the food. It adds water and mucus to the food.

Chemical Digestion:

Saliva also contain an enzyme, salivary amylase, which helps in the semi digestion of staid partially.

Swallowing of the Bolus:

Swallowing is accomplished by muscles movement by the tongue and mouth, food moves into the throat of the pharynx.

Pharynx:

The pharynx is a passage for food and air, is about 5 inches (12.7 centimeters) long thick and muscular part of alimentary canal is present at the posterior end of the oral cavity.

Function:

The bolus from the oral cavity is pushed into the oesophagus through the pharynx. A flexible lap of tissue called the epiglottis reflexively closes over the windpipe which we swallow to prevent choking.

Oesophagus:

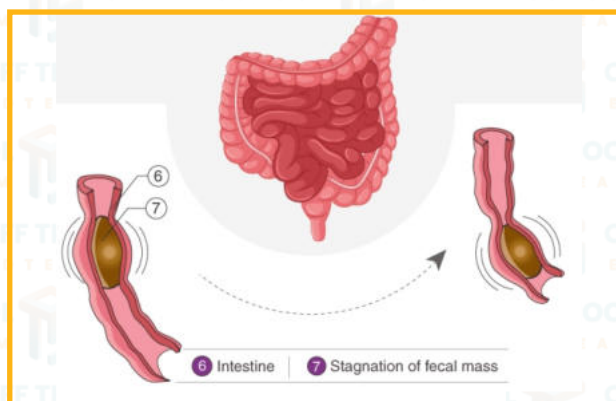
The oesophagus is a muscular tube which connects the pharynx to the stomach.

Function:

Peristalsis is the movement of food from the oral cavity to the rectum.

Peristalsis:

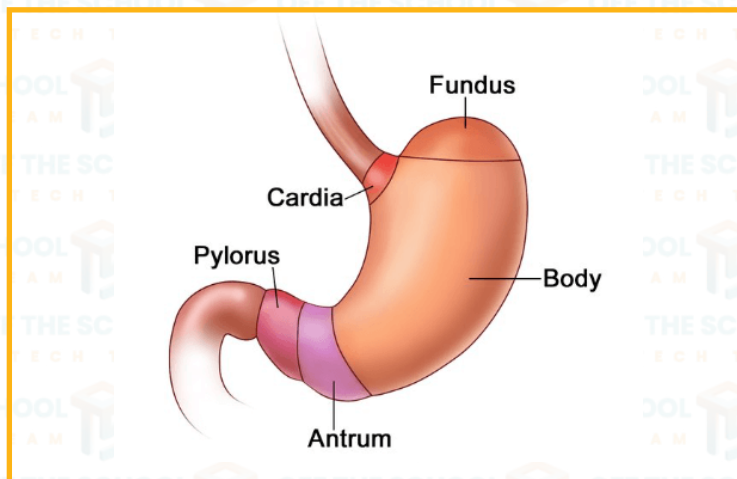
The wave of contraction & relaxation in the smooth muscles of the alimentary canal wall is called Peristalsis



Stomach:

Stomach is a dedicated part of the alimentary canal. It is J-shaped and in the left of the abdomen, just beneath the diaphragm. Stomach has two main portions.

1. Cardiac Portion.
2. Pyloric Portion.

**Cardiac Portion:**

Cardiac portion is present immediately after the oesophagus

Pyloric Portion:

Pyloric portion is located beneath the cardiac end.

Sphincters:

Sphincters are the openings which are guarded by muscles and the stomach has two sphincters.

Pyloric Sphincter:

Cardiac sphincter is between the stomach and oesophagus. Bolus enters stomach from oesophagus through cardiac sphincter

Gastric Juice:

Secretion of gastric gland is known as gastric juice. Gastric juice is composed chiefly of:

- Mucus
- Hydrochloric acid (HCL)
- A protein-digesting enzyme (Pepsinogen)

Function:

- HCL kills the germs & softens the food.
- Enzyme pepsin partly digests the protein into peptones.
- In infants, an enzyme resin helps to turn milk into curds.
- The food is churned up by the contraction and expansion of the stomach.

Small Intestine:

This is a long narrow tube in the form of coded loops. It is differentiated into three distinct parts

1. Duodenum
2. Jejunum
3. Ileum

Duodenum:

It is C-shaped first part of small intestine about 10 inches (25cm) of small intestine where most of the digestive process occurs

Function of Duodenum:

In the small intestine food is further mixed with secretions from the pancreas and liver.

Bile Juice:

Bile juice from the gallbladder first neutralizes the chyme to facilitate digestion and also emulsifies fats.

Pancreatic Juice:

Pancreatic juice from pancreas contains three enzymes.

- Trypsin to digest protein.
- Pancreatic amylase digests carbohydrates
- Lipase digests lipids.

Jejunum:

It is the coiled mid-section about 2.4 meters long after duodenum.

Function:

It is concerned with the rest of the digestion of protein, carbohydrates and lipids.

Ileum:

It is relatively long and more coiled tube which is about 3.6 meters in length and receives completely digested food from jejunum.

Absorption of Food in**Ileum:**

The rest of the digestion is completed in the ileum. The walls of the small intestine house glands which produce intestinal juices; this juice has enzymes aminopeptidase and disaccharidases.

Aminopeptidase:

It converts the peptides into amino acids.

Disaccharides:

Lactose or sucrose is converted into glucose by the action of disaccharidases.

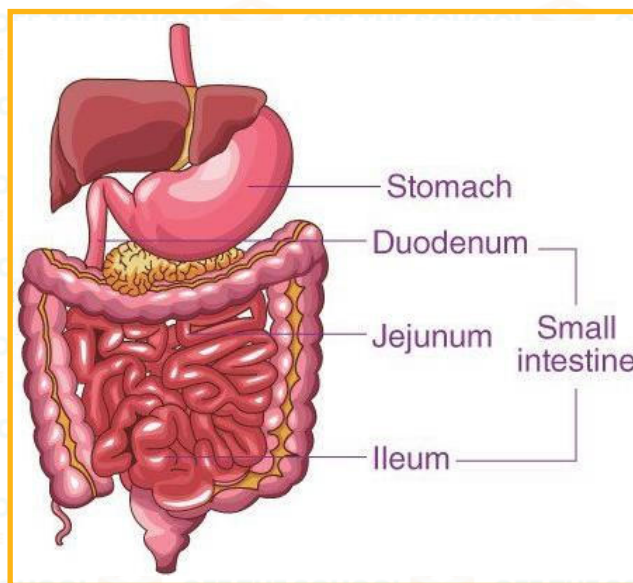
Absorption of Digested Food into the Body Through Villi:

On the internal walls of the ileum there are very fine finger-like projections called villi. Inside each villus there is a rich supply of blood capillaries forming a network. There is also a single lymph vessel called lacteal. Digested food is absorbed in it. Besides fatty acids all the nutrients diffuse through the surface cells of villi and are taken into the blood flowing in blood capillaries. These capillaries join together to form large blood vessels called hepatic portal vein which carries the absorbed food to liver, while fatty acid absorbed by the surface cells of villi rejoin together to form fats which are passed into lacteals of villi join together to form lymph vessels, which form a lymphatic system to deliver these fats into blood stream.

Large Intestine:

The small intestine opens into large intestine which is a comparatively wide tube is subdivided into following regions.

- i. Caecum
- ii. Colon
- iii. Rectum



Caecum:

The caecum is a pouch at the beginning of the large intestine that joins the small intestine to the large intestine. A small hollow, finger like pouch, hangs at the end of the caecum called appendix.

Colon:

The colon extends from the caecum and has three parts.

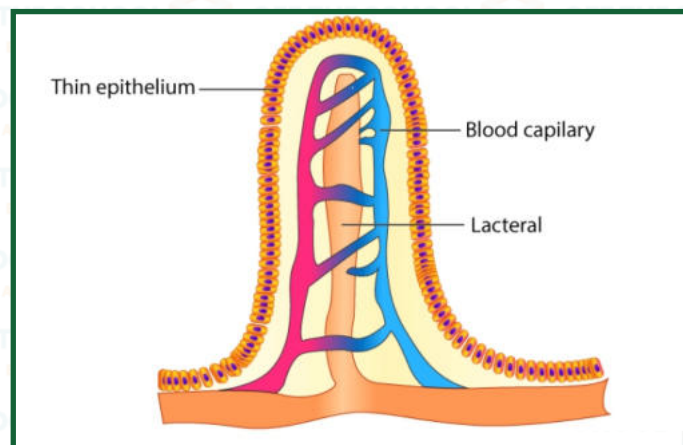
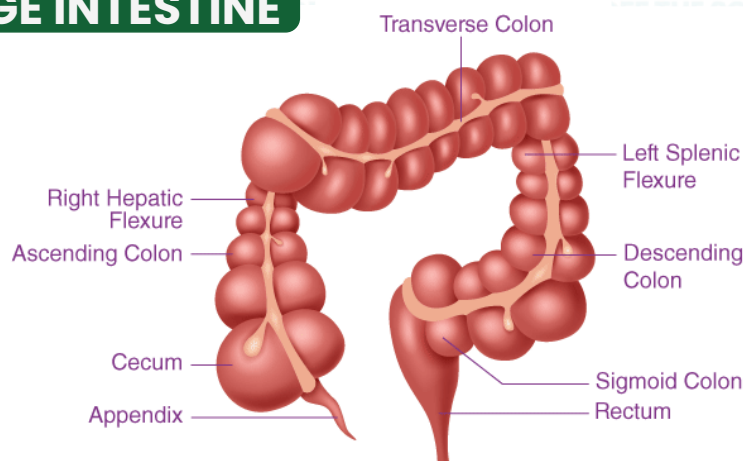
- Ascending: colon absorbs fluids & salts.
- Transverse: colon absorbs fluids & salts.
- Descending: colon, which holds the resulting waste.

Rectum:

Rectum is the terminal part of large intestine where faeces are stored until they leave all digestive systems through the anus.

Functions:

- Inside the ascending & transverse colon, water is reabsorbed from the undigested food into blood.
- Descending colon holds the resulting waste.
- Rectum temporarily stores faeces
- When the rectum is filled up with faeces, it gives rise to reflex anus is opened for defecation.

**LARGE INTESTINE****Digestive Glands:**

In human digestive system secretes digestive chemicals & enzymes which bring out the process of digestion.

- i. Liver.
- ii. Pancreas.

Liver:**Structure:**

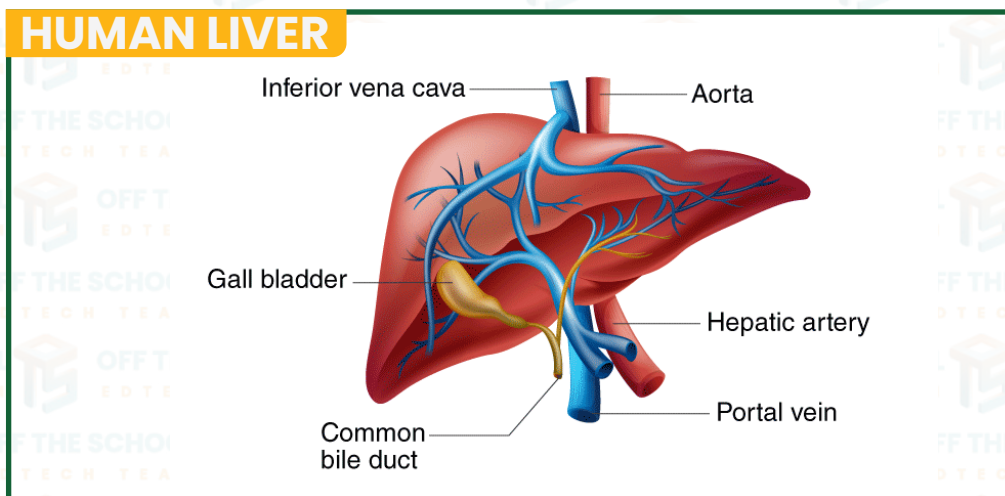
Liver is the largest gland in the body. It is a reddish-brown gland that is located in the abdomen behind the diaphragm more towards the right side.

Functions:

- It secretes an alkaline greenish yellow juice called bile. Bile is stored in a bag-like organ gall-bladder which lies under the surfaces of the liver.
- Bile has no enzymes in it but contains some salts.
- Bile helps in breaking down the larger molecules of fat into smaller particles.
- These smaller fat particles are easy to digest in the small intestine.

Pancreas:**Structure:**

It is a leaf-like organ that lies behind the stomach horizontally within the curve of duodenum.

HUMAN LIVER**Functions:**

- The secretion of pancreas is known as pancreatic juice, which is poured through a duct called pancreatic duct.
- Pancreatic juice contains sodium bicarbonate and many enzymes. Following are the three important enzymes.

Amylase:

It acts on starch to break it down into maltose.

Trypsin:

It converts the proteins into smaller peptides.

Lipase:

It breaks fat droplets into fatty acids and glycerol.

Name of digestive juice	Source	Secretion/Enzyme	Action
Saliva	Salivary glands in oral cavity	Ptyalin	Breaks down starch into maltose
Gastric Juice	Gastric glands in stomach	HCl, Pepsin, Renin	Germicidal activities; pepsin breaks down proteins into peptones, curdles milk

Name of digestive juice	Source	Secretion/Enzyme	Action
Bile	Liver	Nil	Turns acidic chyme alkaline and emulsifies fat
Pancreatic juice	Pancreas	Trypsin	Breaks down proteins into peptides

Disorders of Guts:

Sometimes due to the intake of food, some disorders of the gut or digestive system may develop. Following are some disorders of the gut.

Diarrhea:

It is the condition of large watery motions. It is usually caused by the infection of colon or food poisoning. Continuous conditions of diarrhea could be fatal, because a very large amount of water from the body is lost. So, it is very necessary to rehydrate the person suffering from diarrhea. For rehydration, solution of ORS (oral rehydration salt) must be given continuously to the patient. The physician must be consulted immediately to treat the patient.

Constipation:

Constipation is the condition of frequent or difficult passing of dry hard faces. It is a disorder of the large intestine which absorbs excessive amounts of water from undigested food. It is said to be the mother of all diseases. To avoid constipation a lot of water and food containing more fibers must be taken.

Ulcer:

Peptic ulcer is a sore in the inner wall of the gastrointestinal tract, including the oesophagus, duodenum, or stomach. In ulcer gastric juice gradually breaks down the tissue of the inner wall. Ulcer of the stomach is called gastric ulcer. It is more common in the duodenum than the stomach itself. It can be due to long term use of anti-inflammatory medicine smoking, drinking, coffee, col as spicy food. Few are the reign of abnormal burning after meals, abdominal pain, Rush of saliva after regurgitation, nausea, loss of a peptide & loss of weight. It is treated with medicine which neutralizes the gastric juice, spring acidic food and smoking should be avoided as preventative measures.